Right Triangle Trigonometry & Rationalizing Fractions

Trigonometry Worksheet · Grade 9–11

Name: ________________________________ Date: ___________________

Learning Objectives

- Rationalize fractions by eliminating radicals from the denominator
- Identify the opposite, adjacent, and hypotenuse sides of a right triangle relative to a given reference angle
- Apply the six trigonometric ratios (sin, cos, tan, csc, sec, cot) using SOH-CAH-TOA and its reciprocals

Problems

1. Rationalize the denominator of the fraction shown below.

$$\frac{1}{\sqrt{2}}$$

2. Rationalize the denominator of the fraction shown below.

$$\frac{5}{\sqrt{5}}$$

3. Rationalize the denominator of the fraction shown below.

$$\frac{4}{\sqrt{3}}$$

4. A right triangle has a hypotenuse of 10, an opposite side of 6, and an adjacent side of 8 relative to angle A. Find sin(A) and cos(A).

5. Using the right triangle with opposite = 5, adjacent = 12, and hypotenuse = 13 relative to angle B, find tan(B) and cot(B).

6. For angle C in a right triangle where opposite = 7, adjacent = 24, and hypotenuse = 25, find all six trigonometric ratios.

7. Rationalize the denominator and simplify the expression shown below.

$$\frac{6}{\sqrt{12}}$$

8. A right triangle has angle A at the top-left corner. The side opposite to A has length √3, the adjacent side has length 1, and the hypotenuse has length 2. Find csc(A) and sec(A), then rationalize any radical denominators in your answers.

9. In a right triangle, sin(A) = 3 divided by the square root of 34, and cos(A) = 5 divided by the square root of 34. Rationalize both expressions and then find tan(A) and cot(A).

$$\sin(A) = \frac{3}{\sqrt{34}}, \quad \cos(A) = \frac{5}{\sqrt{34}}$$

10. A right triangle has legs of length 1 and 1, giving a hypotenuse of the square root of 2. Using angle A opposite to one of the legs, find all six trig ratios and rationalize any answers that contain a radical in the denominator.

$$\text{legs} = 1,\ 1; \quad \text{hypotenuse} = \sqrt{2}$$

Right Triangle Trigonometry & Rationalizing Fractions — Answer Key

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Answer Key

1. Answer: $\sqrt{2} / 2$

- Multiply numerator and denominator by $\sqrt{2}$: $(1 \cdot \sqrt{2}) / (\sqrt{2} \cdot \sqrt{2})$
- Denominator becomes $\sqrt{4} = 2$
- Result: $\sqrt{2} / 2$

2. Answer: $\sqrt{5}$

- Multiply numerator and denominator by $\sqrt{5}$: $(5 \cdot \sqrt{5}) / (\sqrt{5} \cdot \sqrt{5})$
- Denominator becomes $\sqrt{25} = 5$
- Result: $5\sqrt{5} / 5 = \sqrt{5}$

3. Answer: $4\sqrt{3} / 3$

- Multiply numerator and denominator by $\sqrt{3}$: $(4 \cdot \sqrt{3}) / (\sqrt{3} \cdot \sqrt{3})$
- Denominator becomes $\sqrt{9} = 3$
- Result: $4\sqrt{3} / 3$ (cannot simplify further)

4. Answer: sin(A) = 3/5, cos(A) = 4/5

- sin(A) = opposite / hypotenuse = 6 / 10 = 3/5
- cos(A) = adjacent / hypotenuse = 8 / 10 = 4/5

5. Answer: tan(B) = 5/12, cot(B) = 12/5

- tan(B) = opposite / adjacent = 5 / 12
- cot(B) is the reciprocal of tan(B) = adjacent / opposite = 12 / 5

6. Answer: sin=7/25, cos=24/25, tan=7/24, csc=25/7, sec=25/24, cot=24/7

- sin(C) = opp/hyp = 7/25
- cos(C) = adj/hyp = 24/25
- tan(C) = opp/adj = 7/24
- csc(C) = hyp/opp = 25/7
- sec(C) = hyp/adj = 25/24
- cot(C) = adj/opp = 24/7

7. Answer: $\sqrt{3}$

- Multiply numerator and denominator by $\sqrt{12}$: $(6\sqrt{12}) / (\sqrt{12} \cdot \sqrt{12})$
- Denominator becomes 12
- Result: $6\sqrt{12} / 12 = \sqrt{12} / 2$
- Simplify $\sqrt{12} = 2\sqrt{3}$, so $(2\sqrt{3}) / 2 = \sqrt{3}$

8. Answer: csc(A) = 2√3/3, sec(A) = 2

- csc(A) = hyp/opp = 2/√3
- Rationalize: (2 · √3) / (√3 · √3) = 2√3 / 3
- sec(A) = hyp/adj = 2/1 = 2 (no rationalization needed)

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9. Answer: sin(A)=3√34/34, cos(A)=5√34/34, tan(A)=3/5, cot(A)=5/3

- Rationalize sin(A): (3 · √34) / (√34 · √34) = 3√34 / 34
- Rationalize cos(A): (5 · √34) / (√34 · √34) = 5√34 / 34
- tan(A) = sin(A)/cos(A) = (3/√34) ÷ (5/√34) = 3/5
- cot(A) = 1/tan(A) = 5/3

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10. Answer: sin=√2/2, cos=√2/2, tan=1, csc=√2, sec=√2, cot=1

- opp = 1, adj = 1, hyp = √2
- sin(A) = 1/√2 → rationalize → √2/2
- cos(A) = 1/√2 → rationalize → √2/2
- tan(A) = 1/1 = 1
- csc(A) = √2/1 = √2
- sec(A) = √2/1 = √2
- cot(A) = 1/1 = 1
